

GROUP 6

NUTRITIONAL PRODUCTS RATER



Daniel Marin Sanchez, Jaime Sanz Villalba, Hannah Steindorfer, Yena Kim

Index

- Introduction
- Data and Methodology
- Rating formula
- Experimental Results
- Discussion
- Conclusion
- Appendices

Introduction

- **Project topic overview**

- Title : “*Nurtritional Products Rater*”
- Objective : To evaluate the nutritional quality of food products and suggest healthier alternatives using Machine Learning methods
- Motivation :
 - With the increasing number of processed foods in modern diets, consumers often struggle to understand how healthy a product really is
 - Nutrition labels can be complex, and marketing claims are often misleading → Our project aims to make this information clear, based on data, and accessible
- By analyzing food data, including nutrients, additives, and categories, we assign a rating score from 0 to 100 that reflects the product’s overall health quality and suggest better alternatives



Introduction



- **Goal**
 - Help users make more informed dietary choices by automatically assessing products through scoring formula and machine learning clustering models
- **Importance of the project**
 - Quantify the nutritional quality of products in a standardized way
 - Identify similar products and recommend healthier alternatives with fewer additives and better nutritional balance
 - Furthermore, contribute to public health awareness and encourage the food industry to improve product transparency and quality

Data and Methodology

QUALITATIVE

- Source category
- Product name
- Barcode
- Categories tags
- Tags of additives



QUANTITATIVE

- Saturated fat / 100 g
- Sugars / 100 g
- Fiber / 100 g
- Proteins / 100 g
- Salt / 100 g
- Energy kcal / 100 g
- Fat / 100 g
- Number of additives
- Rating
- Cluster

1. **Normalize** and **filter** CSV
2. Extract all the **additives** and create **dictionary**
3. **Train** and save the **models** and scalers
4. One model and scaler for each **category**
5. Apply them to the **CSV** in 'Cluster'

Rating Formula

The **Rating Formula** assigns each product a value in the range 0–100 based on two main components:

- **Nutritional Quality (60%)**: Adapted from the Nutri-Score system (energy, sugars, saturated fats, sodium, fiber, and proteins).
- **Additive Impact (40%)**: Penalizes the number and severity of additives using an exponential penalty.
- Goal: **objectivity, interpretability, and scientific justification.**

Q_{nutri} (60%)

- **Negative points A** (0–40):
Energy, Sugars, Saturated fats, Sodium
- **Positive points C** (0–10):
Fiber, Proteins (proteins excluded for beverages)

- **Raw score:** $R = A - C$

- **Normalization:** $Q_{nutri} = \text{clip}_{0,100} \left(100 \cdot \frac{40 - R}{50} \right)$

Information Tables

- Negative points A (0–40):

Points	0	1	2	3	4	5	6	7	8	9	10
Energy (kcal/100g)	< 50	50–80	80–110	110–140	140–170	170–200	200–230	230–270	270–310	310–450	≥ 450
Sugars (g/100g)	< 4.5	4.5–9	9–13.5	13.5–18	18–22.5	22.5–27	27–31	31–36	36–40	40–45	≥ 45
Saturated fats (g/100g)	< 1	1–2	2–3	3–4	4–5	5–6	6–7	7–8	8–9	9–10	≥ 10
Sodium (mg/100g)	< 90	90–180	180–270	270–360	360–450	450–540	540–630	630–720	720–810	810–900	≥ 900

- Positive points C (0–10):

Points	0	1	2	3	4	5
Fiber (g/100g)	< 1	1–2	2–2.8	2.8–3.4	3.4–3.9	≥ 3.9
Protein (g/100g)	< 1.6	1.6–3.2	3.2–4.8	4.8–6.4	6.4–8	≥ 8

Special Case

- But... Since **liquid calories** cause **lower satiety**, **faster absorption**, and **stronger links** to metabolic disease than solid foods, we need **different thresholds** for **beverages**:

Points	0	1	2	3	4	5	6	7	8	9	10
Food (kcal/100g)	< 50	50–80	80–110	110–140	140–170	170–200	200–230	230–270	270–310	310–450	≥ 450
Beverages (kcal/100ml)	< 10	10–20	20–30	30–40	40–50	50–60	60–70	70–80	80–90	90–100	≥ 100

Table Used

- **Official Nutri-Score** scientific document (Sant´e Publique France)
locatable via: **site:santepubliquefrance.fr "scientific and technical report" "nutri-score"** (Includes the full tables of negative and positive points).

Q_add (40%)

- Let **a** denote an **individual additive** in the product, and let **s(a)** be its **severity function**:

$$s(a) = \begin{cases} 0, & \text{non harmful,} \\ 1, & \text{harmful,} \\ 3, & \text{very harmful.} \end{cases}$$

- The **total additive severity** is: $S = \sum_a s(a)$

- The **additive score** is computed as: $Q_{add} = 100 \cdot e^{-\lambda S}, \quad \lambda \approx 0.25$

Final Score Formula

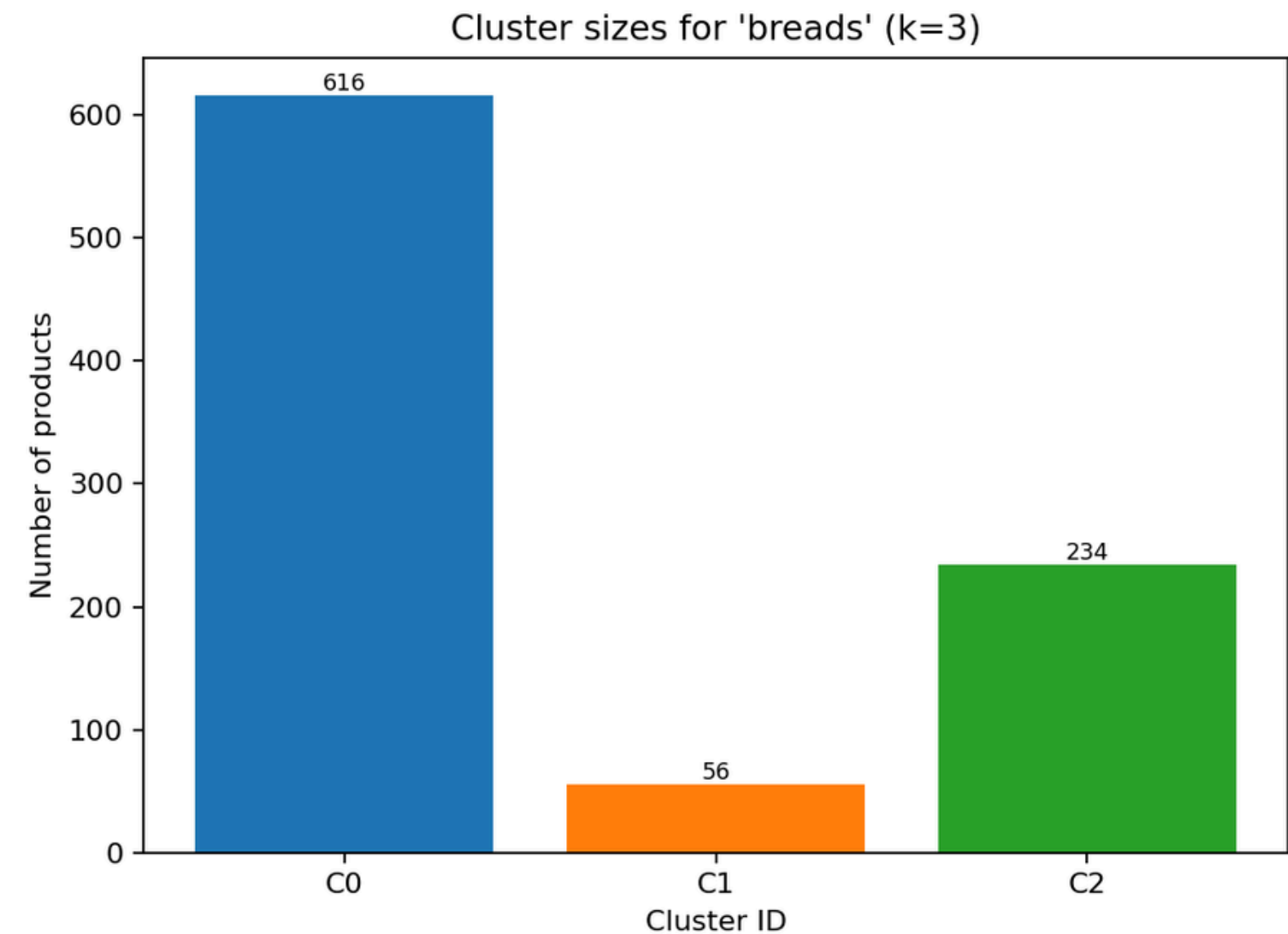
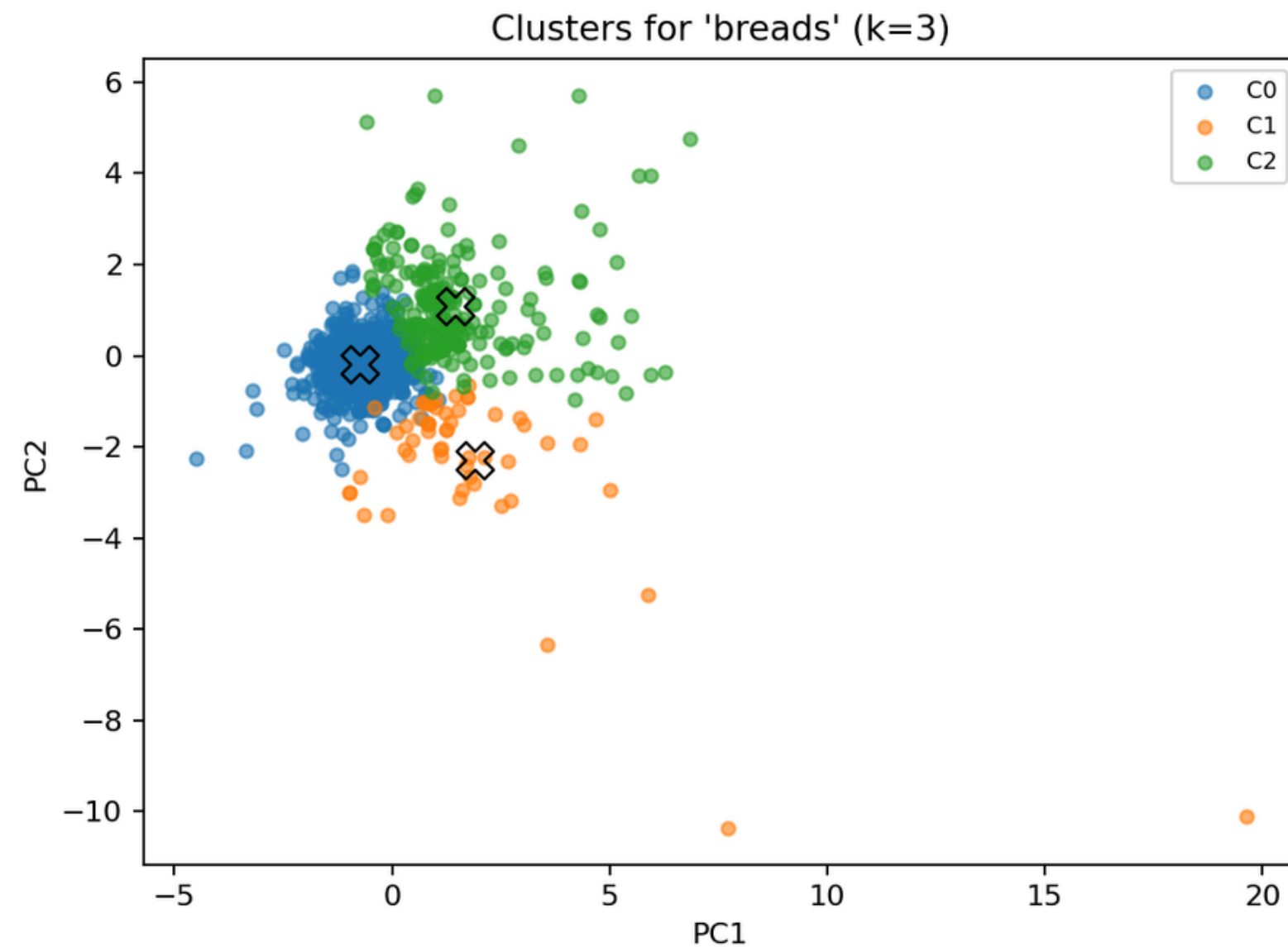
$$\text{Score} = 0.60 \cdot Q_{nutri} + 0.40 \cdot Q_{add}$$

Evaluation

- use silhouette score to decide on best k
- if outliers get assigned their own cluster fall back on elbow score

```
source_category,best_k,criterion,value  
breads,3,silhouette,0.3361217480975359  
breakfast-cereals,2,silhouette,0.30315497628830296  
cheeses,6,elbow_delta,513.3665482382988  
ice-creams,3,silhouette,0.33489633629851556  
meals-with-meat,2,silhouette,0.3499528566894743  
microwave-meals,4,elbow_delta,577.8104690462725  
plant-based-foods,3,elbow_delta,509.7126740291768  
snacks,2,silhouette,0.3724390989377758  
soft-drinks,3,elbow_delta,21.770341860305322  
yogurts,3,silhouette,0.4182808075136468
```

Visualization



Discussion

- **Interpretation of the results**
 - Optimal k values differed widely across categories, indicating that nutritional diversity varies significantly within each group
 - The scoring system successfully distinguished healthier options as alternatives

Discussion

- **Analysis of challenges encountered during the project**
 - inconsistent category and tag data, which initially caused false classifications : bread products misidentified as beverages → required refinement of the parsing logic
 - Nutritional data was incomplete for many items → requiring cleaning and handling strategies

Discussion

- **Future directions**

- Improved Category Classification
 - Implement NLP-based automatic food category classification
 - Enhance accuracy of food vs. beverage detection
- More Accurate Additive Risk Assessment
 - Integrate recent scientific research on additive safety
 - Expand the 'AdditivesDictionary' with more comprehensive additive coverage
- Personalized Scoring
 - Adjust scoring based on individual calorie needs and diet goals
 - Incorporate allergy information and user food preferences

Conclusion

- **Summary of key findings and contributions of the project**
 - Developed an integrated scoring system combining nutritional quality and additive risk into a unified 0–100 health score
 - Created a functional recommendation module that identifies healthier alternatives within the same food category
- **Reflection on the learning experience**
 - Gained experience in handling messy, real world nutritional datasets and developing robust preprocessing strategies
 - Improved skills in Python, data analysis, constructing system, and corporation
 - Learned the importance of precise category detection and the limitations of rule based classification
 - Identified areas for improvement, including more sophisticated recommendation logic and better handling of ambiguous product metadata

Conclusion

- **Final remarks**

- demonstrates how AI systems can support healthier food choices by providing clear, transparent scoring
- The scoring framework can be extended to mobile apps, retail platforms, and personalized dietary guidance tools

The slide features a light gray rectangular area in the center, set against a background with a light beige grid pattern. Olive green geometric shapes are placed in the corners: a triangle in the top-left, a square in the top-right, a square in the bottom-left, and a triangle in the bottom-right. Three thin white vertical lines are positioned on the left side of the bottom-left square, and three thin olive green vertical lines are on the right side of the top-right square.

Q&A